

Complexity Theory

Quiz 3 Set B

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1.5 **mark** for the right answer and $-(.5)$ for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name:

Roll No:

Duration: 40 Minutes

Marks:

1. $P^{Factor} = P$, where $Factor = \{ \langle x, l, u \rangle \mid x \text{ has a factor } y \text{ such that } l \leq y \leq u \}$.

Unknown. $P^{Factor} = P \iff Factor \in P$. If $Factor \in P$, then clearly $P^{Factor} = P$. If $Factor \notin P$, then clearly $P^{Factor} \neq P$ because $Factor \in P^{Factor}$.

2. Recall the stagewise construction of B in the proof of Baker-Gill-Solovay theorem. Let $f(i)$ be the value of n picked in Stage i . Then, f is a strictly increasing function, that is, $\forall i, f(i) < f(i+1)$.

True. If we pick n in Stage i , then membership of at least one string of length n is fixed in Stage i . Hence, in next stage we have to pick a value larger than n .

3. If $NP = coNP$, then $PH = NP$.

True. Hierarchy will collapse to NP .

4. $NP^{SAT} \neq NP$.

False. Since SAT is also in NP , NP^{SAT} is not anymore powerful than NP .

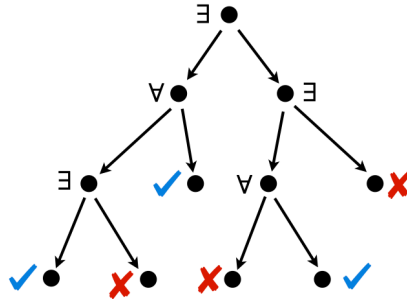
5. $P \subset PH$. ($\subset$ denotes proper subset)

Unknown. If $P \subset PH$, then $P \subset PSPACE$ (because $PH \subseteq PSPACE$) as well which is unknown. If $P = PH$, then $P = NP$.

6. $Path \notin coNL$, where $Path = \{ \langle G, s, t \rangle \mid G \text{ is a directed graph and } s \text{ and } t \text{ are two of its vertices such that there is a path from } s \text{ to } t \}$.

False. Because $Path \in NL$ and $NL = coNL$.

7. Below is the configuration graph of an alternating TM M on an input x .



From the above graph, we can conclude that $x \notin L(M)$.

False.

8. Let SAT_H be the language defined in Ladner's theorem's proof. Then, $SAT_H \in P$.

Unknown. If $SAT_H \in P$, then $H(n) = O(1)$ and we can reduce SAT to SAT_H in polytime and hence, get a polytime algo for SAT . If $SAT_H \notin P$, then $P \neq NP$ as SAT_H is in NP .

9. $TISP(n^{20}, n^5) \not\subseteq \Sigma_2 TIME(n^{15})$. ($\not\subseteq$ denotes not a subset)

False. $TISP(n^{20}, n^5) \subseteq \Sigma_2 TIME(n^{15})$ follows directly from one of the claims in time-space lower bound for SAT .

10. $NP \cup coNP \subseteq \Sigma_2^P \cap \Pi_2^P$.

True. Because NP and $coNP$ are contained in Σ_2^P and Π_2^P .

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1. $\text{NP} \cup \text{coNP} \not\subseteq \Sigma_2^P \cap \Pi_2^P$. ($\not\subseteq$ denotes not a subset)

False.

2. If $\text{NP} = \text{coNP}$, then $\text{PH} = \text{NP}$.

True.

3. Let SAT_H be the language defined in Ladner's theorem's proof. Then, $\text{SAT}_H \in \text{P}$.

Unknown.

4. $\text{NP}^{\text{SAT}} \neq \text{NP}$.

False.

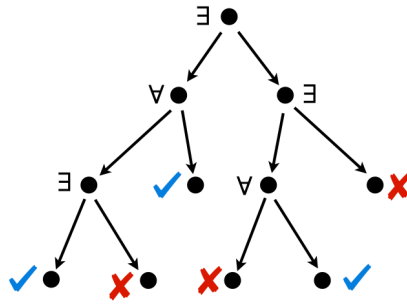
5. $\text{P}^{\text{Factor}} = \text{P}$, where $\text{Factor} = \{ \langle x, l, u \rangle \mid x \text{ has a factor } y \text{ such that } l \leq y \leq u \}$.

Unknown.

6. $\text{TISP}(n^{20}, n^5) \not\subseteq \Sigma_2 \text{TIME}(n^{15})$.

False.

7. Below is the configuration graph of an alternating TM M on an input x .



From the above graph, we can conclude that $x \in L(M)$.

True.

8. Recall the stagewise construction of B in the proof of Baker-Gill-Solovay theorem. Let $f(i)$ be the value of n picked in Stage i . Then, f is an strictly increasing function, that is, $\forall i, f(i) < f(i+1)$.

True.

9. $\text{Path} \notin \text{coNL}$, where $\text{Path} = \{\langle G, s, t \rangle \mid G \text{ is a directed graph and } s \text{ and } t \text{ are two of its vertices such that there is a path from } s \text{ to } t\}$.

False.

10. $P \subset PH$. ($\subset$ denotes proper subset)

Unknown.